Sng Ming Xiang

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PROFILE

Engineering Product Development undergraduate possessing technical competency across the full mechatronics stack, including custom PCB design in KiCad, CAD modeling in Autodesk Fusion 360, and embedded C++ programming for ESP32 microcontrollers. Adept at transforming complex schematic capture into fully functional hardware models, with supplementary experience building basic data analysis scripts in Python.

EDUCATION

Singapore University of Technology and Design (SUTD) — *Singapore*
Bachelor of Engineering (Engineering Product Development) — *Sep 24 - May 28*

- CGPA: 4.0/5.0.

SCHOOL PROJECTS

I Cam'See (Vision-Based Assistive Device) | SUTD Term Project | 2024

- Project introduction: Architected an assistive vision system leveraging a Raspberry Pi 5 single-board computer integrated with hardware camera modules to capture real-time environment data.
- Individual Contribution: Modelled the comprehensive physical site model in Autodesk Fusion 360 to establish structural spatial layouts, subsequently importing the CAD data into Twinmotion to generate high-fidelity animations simulating system workflows.

TagEdu (Digital Education Solution) @ Ho Chi Minh | I&E X UEH GEO | 2025

- Project Introduction: Developed a digital educational platform designed to enhance student engagement and modernise classroom learning experiences through an intuitive interactive solution.
- Individual Contribution: Architected complete user experience workflows by designing high-fidelity wireframes and interactive user interfaces (UI/UX) in Figma, establishing clean data layouts and structural page navigation.

Frankenstein Reimagined (AI Stylometry & Natural Language Processing) | SUTD Project | 2025

- Project Introduction: Evaluated the capability of fine-tuned Large Language Models to replicate complex 19th-century prose syntax and literary styles through an interdisciplinary technical approach.
- Individual Contribution: Fine-tuned a Llama 3.1 LLM using QLoRA and Unsloth to synthesize period-accurate text, subsequently executing statistical data analysis, Principal Component Analysis (PCA), and cluster validation using RStudio.

Ikimono (Emotional Intelligence Wearable) | SUTD Project | 2025

- Project Introduction: Developed an emotional intelligence wearable device designed for children, integrating interactive elements and sensor feedback to track real-time emotion responses.
- Individual Contribution: Engineered the complete IoT implementation, firmware design, and electronics schematic architecture, executing custom PCB design and trace optimization leveraging an ESP32-S3 microcontroller.

TECHNICAL SKILLS

- Electronics & Hardware: PCB Design & Layout, Schematic Capture (KiCad EDA), IoT Architectures, Microcontrollers (ESP32, ESP32-S3), Embedded Systems.
- Design, Modelling & Animation: Computer-Aided Design (CAD), Autodesk Fusion 360, Architectural Rendering & Animation (Twinmotion).
- Software & Data Analysis: Embedded Systems Programming (C++), Scripting (Python), Statistical Validation (R, RStudio).
- AI & Digital Systems: Fine-tuning Large Language Models (LLMs) with QLoRA & Unsloth, UI/UX Wireframing (Figma).
- Version Control: Repository Architecture Management, Collaborative Lifecycle Development (Git, GitHub).

AWARDS & ACTIVITIES

I&E X UEH GEO Competition | 1st Runner Up | 2025

- Awarded 1st Runner Up out of international team entries for the architectural development and presentation of the TagEdu platform.

What the Hack Hackathon | Participant | 2025

- Collaborated in a rapid prototyping hackathon environment to engineer an autonomous moving robotic alarm system.

TOC Taekwondo Open Championships | Participant | 2026

- Competed in the regional Taekwondo open championships; active member of the SUTD Taekwondo club (SUTKD).